

Process, dynamics and bioeffects of acoustic droplet vaporization induced by dual-frequency focused ultrasound

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ABSTRACT

Acoustic droplet vaporization (ADV) plays a crucial role in ultrasound-related biomedical applications. While previous models have examined the stages of nucleation, growth, and oscillation in isolation, which may limit their ability to fully describe the entire ADV process. To address this, our study developed an integrated model that unifies these three stages of ADV, stimulated by a continuous nonlinear dual-frequency ultrasound wave. Using this integrated model, we investigated the influence of nonlinear dual-frequency ultrasound parameters on ADV dynamics and bioeffects by incorporating tissue viscoelasticity through parametric studies. Our results demonstrated that the proposed model accurately captured the entire ADV process, ensuring continuous vapor bubble formation and evolution throughout the phase transition process. Moreover, the applied unified theory for bubble dynamics can simulate intense bubble collapse with high Mach Number as a result of the nonlinear effects of dual-frequency ultrasound. In addition, cavitation-associated mechanical and thermal damage appeared to be more strongly correlated with rapid bubble collapse than with maximum bubble size. Our research also revealed that the mechanical and thermal effects could be regulated independently to some extent by adjusting dual-frequency ultrasound parameters, as they presented differing sensitivities to frequency and acoustic power. Importantly, dual-frequency combinations such as 1.5 MHz + 3 MHz (fundamental and second harmonic), which exhibit a higher Degree of Nonlinearity (DoN) can extend bubble lifespan, offering a potential pathway to the efficacy of ultrasound treatments.

1. Introduction

Acoustic droplet vaporization (ADV) involves the conversion of phase-shift droplets into microbubbles upon exposure to external ultrasound stimulation [1,2]. It has been widely used in biomedical ultrasound applications, including high intensity focused ultrasound (HIFU) ablation [3], boiling histotripsy [4], drug delivery [5], neuro-modulation [6], etc. as a non-invasive and targeted technique. Nano-scale phase-shift droplets have unique advantages in clinical applications due to their longer lifetime and smaller size which can significantly enhance permeability and retention (EPR) effects in vivo [7]. After undergoing a phase shift from liquid to vapor, the newly formed ADV bubbles oscillate under ultrasound pressure, generating stress and heat on the surrounding tissues [8,9]. Understanding the response of nanodroplets to ultrasound waves and their bioeffects on adjacent tissues is critical for the better regulation in ADV applications.

Due to the extremely short duration of the process [10] and the

complexity of the surrounding tissue environment [11], high-speed microscopy studies face significant challenges. As a result, numerical and theoretical research plays a crucial role in understanding the internal mechanisms and influential factors involved in ADV. The entire ADV process can be divided into three continuous stages: nucleation, bubble growth and recondensation, and subsequent bubble oscillation [12], based on the phase behavior of acoustically sensitive liquids such as perfluorocarbon (PFC) [13]. These stages are closely interconnected, with each stage having a substantial impact on the following one [12,14]. However, previous numerical studies often focus on individual stages in isolation, neglecting the connections between them. In these studies, the initial conditions of each stage are treated as independent of the preceding stage. For example, in Ref [15], Ghasemi et al. fixed the initial radius of the bubble as 80 nm to study the effects of acoustic amplitude on ADV process. And Lacour et al. [16] also used similar assumptions and methods. In fact, considering the continuity of the ADV process, the initial radius of the bubble inside the droplet is also

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influenced by acoustic amplitude [17] and should not be set as fixed. Additionally, external environmental factors, droplet properties, and driving ultrasonic conditions are often inconsistent. Therefore, the primary objective of our work is to develop a unified, whole-process ADV model that integrates nucleation, growth and recondensation, and oscillation stages, while ensuring consistency in external environmental parameters. This can facilitate a more comprehensive understanding of the process and underlying mechanisms of ADV, thereby enabling precise modulation of this process during ultrasound treatment, ultimately enhancing therapeutic efficacy.

It has been reported that, compared to standard single-frequency ultrasound, dual-frequency ultrasound excitation can be used to control nanodroplet vaporization and enhance the subsequent inertial cavitation activity of the vapor bubble [18–20]. However, the influence of dual-frequency ultrasound parameters, including frequency combinations and acoustic powers, on the entire ADV process remains unclear. To address this, it is necessary to extend our model to accommodate dual-frequency ultrasound conditions. In such cases, the effects of ultrasonic nonlinearity [21,22], such as waveform distortion and high harmonics, must be considered. These nonlinear effects result in more severe ADV bubble dynamics activities than those observed under traditional single-frequency excitation [23], posing a challenge for current cavitation bubble dynamics models.

Currently, several bubble dynamics models exist, including Rayleigh-Plesset model [24], Herring-Trilling model [25], Keller-Miksis model [26] and Gilmore model [27]. Due to its relatively wide applicability [28], the Gilmore model has been the most frequently used in recent studies of acoustic cavitation bubble dynamics among these models. However, the basic assumptions underlying the Gilmore model impose a strict limitation: the Mach Number at bubble collapse must remain below 2.2 during simulation (It should be pointed out that the Mach Number mentioned in this article refers to its absolute value.) [27]. In contrast, other models impose more stringent constraints on the Mach Number. For instance, the Keller-Miksis model specifies a Mach Number threshold of less than 1, and the Herring-Trilling model specifies a Mach Number threshold of less than 0.5. While this restriction is appropriate in most cases, dual-frequency nonlinearity may render it unsuitable for the current study. Related experimental results have shown that the Mach Number at bubble collapse can exceed 4 under certain conditions [29]. Therefore, a new bubble dynamics model is required to simulate bubble oscillation under nonlinear dual-frequency ultrasound excitation. In a recent study, Zhang et al. [30] developed a novel theory for the dynamics of oscillating bubbles, which unifies various classical bubble equations. This unified theory eliminates the limitation that the Mach Number must remain below 2.2, which demonstrates significant advantages over existing models. This theory has been extensively applied in simulations of cavitation bubble hydrodynamics, accounting for complex phenomena such as jet formation and shock wave dynamics [31,32]. Therefore, the unified theory for bubble dynamics has been applied in this work to simulate the intense bubble collapse stimulated by nonlinear dual-frequency ultrasound excitation.

Most previous studies on the ADV process have assumed that the surrounding medium behaves as a Newtonian fluid. For instance, Doi-nikov et al. [33] developed a theoretical model describing the growth and subsequent oscillation of a vapor bubble inside the droplet, based on a concentric three-phase physical model (vapor bubble – PFC droplet – host liquid). However, in practical applications, it is inappropriate to assume biological tissues behave as Newtonian fluids due to their elastic properties [34]. Recently, Feng et al. [35] developed a vaporization model to study the effects of tissue viscoelasticity and adjacent micro-bubbles on the vaporization process and the direct growth threshold. In their work, the Kelvin-Voigt model was used to represent the ambient viscoelastic tissue. Given the extremely short bubble growth time, the relaxation process of the tissue cannot be ignored [36], as it is comparable to the period of the ultrasound wave. Most ultrasound durations for ADV are microseconds (μs), given that the frequency typically

applied is in the MHz range. It is reported that the relaxation time of the viscoelastic biological medium has a wide range of $10^{-11} - 10^{-7}$ s [37,38]. Therefore, the Zener model [38], which accounts for relaxation, should be employed for a more precise prediction of tissue viscoelastic behavior.

The classical nucleation theory (CNT), previously applied to study cavitation in biological fluids and tissues [39] and nucleation in nano- and micro-sized phase-shift droplet emulsions [40], is utilized here to simulate acoustic cavitation. CNT has been employed to estimate the ADV threshold [41] and the condensation threshold of ADV bubbles [42]. In our previous work [43], by appropriately adjusting the surface tension of the liquid, the modified CNT has been demonstrated to effectively simulate the ADV nucleation of PFP nanodroplets activated by nonlinear dual-frequency ultrasound.

To summarize, this study aims to develop a unified whole-process model that integrates nucleation, growth and recondensation, and oscillation, while investigating its dependency on nonlinear dual-frequency ultrasound, considering tissue viscoelasticity. The nucleation stage was modeled using a modified CNT [17]. A numerical model, based on fluid dynamics and thermodynamics, was developed to simulate vapor bubble growth and recondensation inside the nanodroplet. After vaporization, the oscillation of the vapor bubble was described using the unified theory of bubble dynamics. The Zener viscoelastic model was employed to simulate the properties of soft tissue. The entire process is continuous and driven by the ultrasound waves obtained from time-domain nonlinear dual-frequency ultrasound simulations. Quantitative studies were conducted on the acoustic cavitation bubble dynamics and the spatiotemporal distribution of mechanical stress and heat deposition in the surrounding tissues. Additionally, parametric studies were performed to investigate the effects of dual-frequency focused ultrasound parameters on ADV and its biomedical effects.

2. Theory and methods

2.1. Overall of the whole-process vaporization dynamics model

Fig. 1 provides a schematic representation of the whole-process vaporization dynamics of a nanodroplet inside the viscoelastic tissue. The acoustic-sensitive PFP nanodroplets were assumed to be uniformly distributed throughout the tissue phantom which would be activated to microbubbles under the excitation of the dual-frequency focused ultrasound. As the initial stage of phase-transition process, a critical nucleus would be formed inside the nanodroplet firstly, which is called as nucleation. At this point, the liquid–gas mixture is unstable and proper acoustic waves could help vapor grow to resist the natural condensation [44]. When the phase transformation of PFP nanodroplet is completed, the vapor bubble will experience oscillatory stage. In this system, three types of media are present, denoted by the subscripts 1, 2, and 3, which correspond to the vapor bubble, the liquid PFP, and the viscoelastic tissue, respectively.

2.2. ADV nucleation model for a PFP nanodroplet activated by dual-frequency ultrasound

A modified CNT [17] was employed to model the ADV nucleation of a PFP nanodroplet activated by dual-frequency ultrasound and the radius of the critical nucleus r^* and the logarithmic form of the nucleation rate J can be calculated by:

$$\begin{cases} r^*(T_{2n}, p_2) = \frac{2\sigma_2(T_{2n}, p_2)}{p_v - p_2} \\ \lg J(T_{2n}, p_2) = \lg_{10} \left(\sqrt{\frac{3\sigma_2(T_{2n}, p_2)p_2^2}{\pi m^3}} \cdot \exp \left(-\frac{W^*(T_{2n}, p_2)}{k_B T_{2n}} \right) \right) \end{cases} \quad (1)$$

where σ_2 is the size-dependent surface tension of PFP liquid and it is a

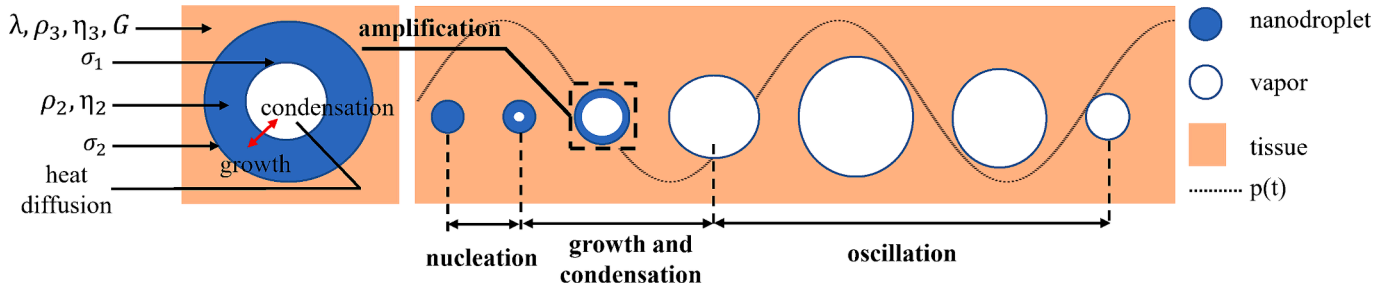


Fig. 1. Schematic diagram of the whole-process vaporization dynamics of a nanodroplet in viscoelastic tissue.

function of temperature T_n and pressure p_2 considering the effect of the nucleus curvature based on a scaling function. p_v is the vapor pressure in the nucleus. ρ_2 represents the density of PFP liquid, m represents the mass of a single PFP molecule, k_B is the Boltzmann constant, and W^* represents the critical work for nucleation. Further detailed information on the ADV nucleation model can be found in our previous work [43,45].

The nucleation rate J is defined as the nuclei formed in the unit time (1 s) and volume (1 m^3) which is significantly larger than that in practical ADV applications. Therefore, another parameter called the number of critical nuclei N_J was introduced which can be calculated by $\lg N_J = \log_{10}(J^* V_0^* V_f^* \tau_0)$. Here, $V_0 = 1 \times 10^{-6} \text{ mm}^3$ and τ_0 is calculated as $\tau_0 = 1/(10f)$ [39]. Finally, the nucleation modeled is completed when N_J equals to 1 and the corresponding critical nuclei size is used as the initial value of the vapor bubble radius inside the phase-shift droplet.

2.3. Vapor bubble growth and recondensation model inside an acoustically activated PFP nanodroplet

To model the growth and recondensation of a vapor bubble nucleated in the center of a PFP nanodroplet, a three-phase computational domain was considered which has been shown in the left of Fig. 1 with amplification. A spherical vapor bubble with a radius of $R_1(t)$ is located at the center of a spherical PFP nanodroplet with a radius of $R_2(t)$ which is surrounded by the viscoelastic tissue.

For the sake of simplicity, it is assumed that the system is symmetric and that droplet and vapor preserve their spherical shape during the simulation. In addition, the gas inside the bubble is an ideal gas and the pressure and temperature inside the vapor bubble is spatially uniform. Moreover, these media are incompressible viscous liquids which obey the equation of continuity and the Navier-Stokes equation and the density of the nanodroplet ρ_2 and the density of the tissue ρ_3 are constant [46,47].

In a one-dimensional spherical coordinate system, the conservation of mass equation can be given as [48]:

$$\begin{cases} r^2 \nu_s = R_1^2 \left(\dot{R}_1 - \frac{\xi}{\rho_1} \right) = R_1^2 \left(\dot{R}_1 - \frac{\xi}{\rho_2} \right) = R_2^2 \dot{R}_2, & \text{for } R_1 \leq r \leq R_2 \\ r^2 \nu_s = R_2^2 \dot{R}_2, & \text{for } R_2 < r \end{cases} \quad (2)$$

where the subscript $s = 2$ or 3 denotes the liquid nanodroplet or the tissue, respectively, r is the radial coordinate, ν stands for velocity, the dot means time derivatives. $\nu_1 = \dot{R}_1 - \xi/\rho_1$ and $\nu_2 = \dot{R}_1 - \xi/\rho_2$ which could be easily gotten from Eq. (2) at $r = R_1(t)$. ρ_1 is the density of the vapor bubble and ξ is the evaporation rate. It is important to note that during bubble growth and recondensation, only a phase transition occurs between PFP liquid and gas, with no other gases, such as water vapor, entering or leaving the PFP bubbles. This assumption simplifies the model and improves the convergence of calculations. This assumption has been adopted in previous related studies [33,35].

In a one-dimensional spherical coordinate system, based on the assumption of incompressibility of the media, the conservation of mo-

mentum equation can be given as [49]:

$$\rho_s \left(\frac{\partial \nu_s}{\partial t} + \nu_s \frac{\partial \nu_s}{\partial r} \right) = - \frac{\partial p_s}{\partial r} + \frac{\partial \tau_{s,rr}}{\partial r} + \frac{3\tau_{s,rr}}{r} \quad (3)$$

where $\tau_{s,rr}$ is component of the stress tensor in the r direction. By substituting Eq. (2) into Eq. (3) and then integrating it from r to ∞ , we can get a second-order ordinary differential equation governing the radial motion of the nanodroplet:

$$\begin{aligned} & \rho_2 R_1 \dot{\nu}_2 \left(1 + \frac{\rho_3 - \rho_2}{\rho_2} \frac{R_1}{R_2} \right) + \frac{3}{2} \rho_2 \dot{R}_1 \nu_2 \left[1 + \frac{\rho_3 - \rho_2}{\rho_2} \frac{R_1}{R_2} \left(\frac{4R_2^3 - R_1^3}{3R_2^3} \right) \right] \\ & - \frac{\nu_2 \xi}{2} \left(1 + \frac{\rho_3 - \rho_2}{\rho_2} \frac{R_1^4}{R_2^4} \right) \\ & = p_2(R_1, t) - p_2(R_2, t) + p_3(R_2, t) - p_\infty(t) + \tau_{2,rr}(R_2, t) \\ & - \tau_{2,rr}(R_1, t) - \tau_{3,rr}(R_2, t) + 3 \int_{R_1}^{R_2} \frac{\tau_{2,rr}}{r} dr + 3 \int_{R_2}^{\infty} \frac{\tau_{3,rr}}{r} dr \end{aligned} \quad (4)$$

where p_2 , p_3 and $p_\infty(t) = p_0 + p_{ac}(t)$ are the pressures in the nanodroplet, the tissue, and at infinity, respectively and $p_{ac}(t)$ is the driving pressure obtained from the acoustic field simulations.

2.4. Bubble oscillation model in the viscoelastic tissue

When the PFP nanodroplet completely turns into a vapor bubble, the oscillation of the bubble is described by the unified theory for bubble dynamics [30]:

$$R_1 \ddot{R}_1 \left(1 - \frac{\dot{R}_1}{C} \right) + \frac{3}{2} \dot{R}_1^2 \left(1 - \frac{\dot{R}_1}{3C} \right) = \left(1 + \frac{\dot{R}_1}{C} \right) H + \frac{R_1}{C} \dot{H} \quad (5)$$

where H and C denote the values of enthalpy and sound speed at the bubble wall respectively. They can be calculated as follows:

$$H = \frac{1}{\rho_3} \frac{n}{n-1} (p_\infty + B) \left(\left(\frac{p_3 + B}{p_\infty + B} \right)^{\frac{n-1}{n}} - 1 \right) \quad (6)$$

$$C = c_3 \left(\frac{p_3 + B}{p_\infty + B} \right)^{\frac{n-1}{2n}} \quad (7)$$

where $B = c_3^2 \rho_3 / n - p_0$, and $n = 7$ are particular liquid constants [50].

Considering that the bubbles are in the viscoelastic tissue, Eq. (5) can be written as [27,51]:

$$\begin{aligned} & R_1 \ddot{R}_1 \left(1 - \frac{\dot{R}_1}{C} \right) + \frac{3}{2} \dot{R}_1^2 \left(1 - \frac{\dot{R}_1}{3C} \right) = \\ & \left(1 + \frac{\dot{R}_1}{C} \right) \left(H + \frac{1}{\rho_3} \int_{R_1}^{\infty} \left(\frac{\partial \tau_{3,rr}}{\partial r} + \frac{3\tau_{3,rr}}{r} \right) dr \right) \\ & + \frac{R_1}{C} \left(\dot{H} + \frac{d}{dt} \left(\frac{1}{\rho_3} \int_{R_1}^{\infty} \left(\frac{\partial \tau_{3,rr}}{\partial r} + \frac{3\tau_{3,rr}}{r} \right) dr \right) \right) \end{aligned} \quad (8)$$

2.5. Initial condition and boundary condition of the model

In the whole-process vaporization dynamics model presented in this paper, 3 different but successive stages including nucleation, growth and recondensation and oscillation have been integrated. Sometimes the same physical quantity follows different computational formulas in different stages. In this section, detailed description of the physical quantities which have been introduced above and how these stages are connected will be provided.

When nucleation, T_{2n} and p_2 are determined by the temperature and pressure fields of dual-frequency focused ultrasound. The nonlinear acoustic field is described by the generalized Westervelt equation [52] and the temperature in the tissue phantom produced by the ultrasound thermal effect was calculated by the Pennes' Bioheat Transfer equation [53]. For Eq. (1), T_{2n} and p_2 are determined when the number of critical nuclei $N_J = 1$, which means only 1 critical nucleus is formed. For more information on N_J , please refer to our previous work [43,45]. In addition, it also ensures that there is only one bubble and interactions between bubbles are ignored here. The r^* will be the initial radius of the vapor bubble R_1 and T_{2n} will be the initial value of T_2 in the growth and recondensation stage.

Before nucleation, there are only two types of mediums: PFP nanodroplet and tissue. Once a critical nucleus is formed, the third type of mediums (i.e. PFP vapor) is produced. According to the equation of state of ideal gas, the density ρ_1 , pressure p_1 and temperature T_1 of vapor can be described by the following formula [48,54]:

$$\rho_1 = \frac{p_1}{R_{g1} T_1} \quad (9)$$

$$T_1 = \left[\frac{1}{T_{s0}} - \frac{R_{g1}}{L} \ln \left(\frac{p_1}{p_0} \right) \right]^{-1} \quad (10)$$

$$R_1 \dot{p}_1 + 3\gamma_1 p_1 \left(\dot{R}_1 - \frac{\xi}{\rho_1} \right) = 0 \quad (11)$$

where $R_{g1} = 28.86 \text{ J}/(\text{mol}\cdot\text{K})$ is the specific gas constant of the vapor bubble, $T_{s0} = 29^\circ\text{C}$ is the saturation temperature at the reference pressure $p_0 = 0.1013 \text{ MPa}$ and $L = 1.168 \times 10^5 \text{ J/kg}$ is the latent heat of evaporation. The evaporation rate ξ can be obtained from the conservation of energy at the interface, which is proportional to the heat transfer flux on the nanodroplet side [55]:

$$\xi = \frac{\kappa_2}{L} \left(\frac{\partial T_2}{\partial r} \right) \text{ at } r = R_1 \quad (12)$$

where κ_2 is the thermal conductivity of the nanodroplet, T_2 is the temperature field inside the nanodroplet. $T_2(r, t)$ is described according to the conservation of energy equation in a one-dimensional spherical coordinate system as follows [54]:

$$\frac{\partial T_2}{\partial t} + v_2 \frac{\partial T_2}{\partial r} - \frac{1}{r^2} \frac{\partial}{\partial r} \left(\chi_2 r^2 \frac{\partial T_2}{\partial r} \right) = \frac{12\eta_2}{\rho_2 c_2} \frac{v_2^2}{r^2} \quad (13)$$

where η_2 is the viscosity and ε_2 is the specific heat of the nanodroplet. $\chi_2 = \kappa_2/(\rho_2 c_2)$ is the thermal diffusivity of the nanodroplet. The initial condition and boundary condition for Eq. (13) is as follows:

$$\begin{cases} T_2(r, t) = T_1(r, t), & \text{at } r = R_1 \\ T_2(r, t) = T_{2n}, & \text{at } r \rightarrow \text{nanodroplet - tissue interfaces} \\ T_2(r, t) = T_{2n}, & \text{at } t = 0 \end{cases} \quad (14)$$

Eq. (4) describes the growth of vapor bubble inside a PFP nanodroplet and it contains several quantities to be determined. The nanodroplet radius R_2 can be obtained from the condition of conservation of mass to the nanodroplet [46]:

$$\dot{R}_2 = \frac{R_1^2}{R_2^2} \left(\dot{R}_1 - \frac{\xi}{\rho_2} \right) \quad (15)$$

According to the continuity of normal stress including surface tension effects and scattering effect, the pressures at the two interfaces are given by [35]:

$$p_1(R_1, t) = p_2(R_1, t) + \frac{2\sigma_1}{R_1} - \tau_{2,rr}(R_1, t) + \xi(\nu_2 - \nu_1) \text{ at } r = R_1 \quad (16)$$

$$p_2(R_2, t) - \tau_{2,rr}(R_2, t) = p_3(R_2, t) - \tau_{3,rr}(R_2, t) + \frac{2\sigma_2}{R_2} \text{ at } r = R_2 \quad (17)$$

The liquid nanodroplet behaves as Newtonian fluids, so we can get [49]:

$$\tau_{2,rr} = 2\eta_2 \left(\frac{\partial \nu_2}{\partial r} \right) \quad (18)$$

Zener model has been used in this paper to simulate the viscoelastic tissue thus we have [38]:

$$\tau_{3,rr} + \lambda_3 \dot{\tau}_{3,rr} = 2G\gamma_{3,rr} + 2\eta_3 \dot{\gamma}_{3,rr} \quad (19)$$

where $\gamma_{3,rr}$ is a strain and $\dot{\gamma}_{3,rr}$ is strain rate and they turn out to be:

$$\gamma_{3,rr} = -\frac{2}{3r^3} (R_s^3 - R_{20}^3) \quad (20)$$

$$\dot{\gamma}_{3,rr} = -2 \frac{R_s^2 \dot{R}}{r^3} \quad (21)$$

By introducing a new variable $q = \int_{R_s}^{\infty} \frac{\tau_{3,rr}(r, t)}{r} dr$, dividing Eq. (19) by r and integrating the resulting equation from R_s to ∞ , we can get [32]:

$$q + \lambda_3 \dot{q} + \lambda_3 \frac{\dot{R}_s \tau_{s,rr}|_{R_s}}{R_s} = \frac{1}{3} \left(-\frac{4G}{3R_s^3} (R_s^3 - R_{20}^3) - 4\eta_3 \frac{\dot{R}_s}{R_s} \right) \quad (22)$$

and $\tau_{s,rr}|_{R_s}$ could also be obtained from Eq. (19) at $r = R_s$:

$$\tau_{s,rr}|_{R_s} + \lambda_3 \dot{\tau}_{s,rr}|_{R_s} = -\frac{4G}{3} \left(1 - \frac{R_{20}^3}{R_s^3} \right) - 4\eta_3 \frac{\dot{R}_s}{R_s} \quad (23)$$

It should be noted that in Eqs. (20) – (23), the subscript $s = 2$ is used in growth and recondensation stage while $s = 1$ is used in oscillation stage because the media subjected to tissue stress are different in different stages.

Substituting Eqs. (15) – (23) into Eq. (4), a motion equation governed the vaporization process of the nanodroplet in a viscoelastic tissue can be obtained which can be expressed as:

$$\begin{aligned} \rho_2 R_1 \dot{\nu}_2 \left(1 + \frac{\rho_3 - \rho_2}{\rho_2} \frac{R_1}{R_2} \right) + \frac{3}{2} \rho_2 \dot{R}_1 \nu_2 \left[1 + \frac{\rho_3 - \rho_2}{\rho_2} \frac{R_1}{R_2} \left(\frac{4R_2^3 - R_1^3}{3R_2^3} \right) \right] \\ + \frac{\xi}{3} \left(1 + \frac{\rho_3 - \rho_2}{\rho_2} \frac{R_1^4}{R_2^4} \right) = p_1 - p_0 - p_{ac}(t) \\ - \xi^2 \left(\frac{1}{\rho_1} - \frac{1}{\rho_2} \right) - \frac{2\sigma_2}{R_2} - \frac{2\sigma_1}{R_1} - 4 \frac{\left(\dot{R}_1 - \xi \right)}{R_1} \left[\eta_2 \left(\left(1 - \eta_2 \right) \frac{R_1^3}{R_2^3} \right) \right] + 3q \end{aligned} \quad (24)$$

Substituting Eqs. (19) – (23) into Eq. (8), the bubble oscillation model in the viscoelastic tissue can be described as:

$$R_1 \ddot{R}_1 \left(1 - \frac{\dot{R}_1}{C}\right) + \frac{3}{2} \dot{R}_1^2 \left(1 - \frac{\dot{R}_1}{3C}\right) = \left(1 + \frac{\dot{R}_1}{C}\right) \left(H + \frac{1}{\rho_3} (3q - \tau_{3,rr}|_{R_1})\right) + \frac{R_1}{C} \left(\dot{H} + \frac{1}{\rho_3} (3\dot{q} - \tau_{3,rr}|_{R_1})\right) \quad (25)$$

When the PFP droplet completely turns into a vapor bubble, the system immediately enters the next oscillation stage. The values of R_1 and $\dot{R}_1$ at the end of vaporization process will be the initial values of that in oscillation stage. At this moment, it returns to a system of two types of media: vapor and tissue. So Eq. (16) and Eq. (17) can be rewritten as:

$$p_3 = p_1 - \frac{2\sigma_1}{R_1} + \tau_{3,rr}|_{R_1} \quad (26)$$

It should be pointed out that p_1 in Eq. (26) here is no longer follow the calculation method describe in the growth stage in Eqs. (9) – (11). In the growth and recondensation stage, the mass of vapor is increasing because the droplet continues to vaporize, however, in the oscillation stage, the mass of vapor is fixed and the pressure inside the bubble p_1 is assumed to follow adiabatic equation of state [56]:

$$p_1 = \left(p_0 + \frac{2\sigma_1}{R_{10}}\right) \left(\frac{R_1}{R_{10}}\right)^{3\gamma} \quad (27)$$

where R_{10} is the initial value of R_1 in the oscillation stage, i.e. the value of R_1 at the end of vaporization process and $\gamma = 1.095$ is the polytropic exponent.

2.6. Heat deposition during the whole process

During the acoustic droplet vaporization, the motion of the bubble and droplet will produce heat deposition to the tissue. There are three types of heat deposition caused by the cavitation [57]. The first is the ultrasound power deposition term Q_{ac} and $Q_{ac} = \frac{\alpha p_{ac}^2}{\rho_3 c_3}$. It is caused by the absorption of ultrasound by tissue where $\alpha = 8.55 \text{ Np/MHz/m}$ is the attenuation coefficient of the tissue. The remaining two terms are caused by the cavitation related thermal effects of the radiated pressure wave. The viscous damping term is denoted as Q_{vis} and the spatial averaged power density of it is defined as follows:

$$Q_{vis} = \frac{16\pi\eta_3 R_s \dot{R}_s^2}{\frac{4}{3}\pi r^3} \quad (28)$$

where the subscript s follows the way used in Eqs. (20) – (23) and r' is the radius of the volume for spatial average, upon which ultrasound power is deposited.

Once the bubble is formed, it could radiate ultrasound outward and the radiated pressure P_{rad} at the distance r from the bubble surface can be calculated as follows:

$$P_{rad} = \frac{\rho_3 R_1}{r} (2\dot{R}_1^2 + R_1 \ddot{R}_1) \quad (29)$$

And the power deposited through a sphere with radius r' is defined as follows:

$$D_{rad}^{abs} = 4\pi r'^2 \frac{P_{rad}^2}{\rho_3 c_3} (1 - e^{-2\alpha r'}) \quad (30)$$

Finally, the last term of heat deposition which is responsible for the absorption of the radiated pressure wave can be calculated as follows by performing the spatial averaging within a sphere of radius r' :

$$Q_{rad} = \frac{3}{4\pi r'^3} D_{rad}^{abs} \quad (31)$$

2.7. Simulation conditions and data analysis

Parametric studies were conducted to investigate the effects of dual-frequency ultrasound on the whole-process vaporization dynamics of a PFP nanodroplet in the viscoelastic tissue. The parameters of dual-frequency ultrasound are the same as that used in our previous study [45]. For low frequency ultrasound, three different driving frequencies (LF = 0.5, 1.0, and 1.5 MHz) and eight different acoustic powers (LW ranging from 0 to 70 W with increment of 10 W) were selected and for high frequency ultrasound, we identified a set of parameters with driving frequencies between 3 MHz and 5 MHz, increasing in 1 MHz increments (HF = 3, 4, and 5 MHz), and acoustic powers ranging from 0 W to 25 W, increasing by 5 W intervals. The nonlinear propagation of dual-frequency ultrasound is simulated by an open-source MATLAB toolbox called k-Wave Version 1.4 [58]. The waveforms obtained from the simulations are used as the driving pressure for inducing ADV. This part of the work has already been reported in our previous article [45] and the results of peak positive and negative pressure under different dual-frequency focused ultrasound conditions of nonlinear model are shown in the [Supplementary Material](#).

Unless specified otherwise, the physical properties of the medium and other parameters used in the simulations are listed in Table 1. The tissue-mimicking phantom used in this work did not refer to a specific type of human tissue, but was set as a uniform medium with average values of physical properties for human soft tissue. The values of the physical properties of the medium are determined referenced to Refs [35,51,56].

The partial differential equation Eq. (13) is solved by being divided into a system of ordinary differential equations (ODEs) with second-order finite differences [59]. However, this approach may lead to convergence issues during the solution process. Consequently, we must establish a criterion for the vaporization process and its completion. Since we need to enter the oscillation stage immediately, we aim for the growth process to be as fully transformed as possible. Consequently, we keep calculating until $R_1 = 99.9999999\%$ of R_2 and it is a strict criteria which extends the duration of bubble expansion driven by ultrasound negative pressure. The radius of the bubble at the completion of the growth process will be relatively large, but this does not violate the law of conservation of mass. As the bubble grows, its density decreases while its total mass remains constant. The ODEs were solved numerically using the ODE15s solver built in MATLAB.

3. Results

3.1. Verification and validation

In order to justify the employed simulation model, some verification and validation studies were conducted. Fig. 2 shows the results of the comparisons between the results of our model and previous numerical models. A one-period pulse, $p_{ac}(t) = -A\cos(2\pi ft)$, was used as acoustic

Table 1
Simulation parameters.

Symbol	Definition	Values	Units
σ_1	surface tension at PFP vapor-droplet interface	0.00854	N/m
σ_2	surface tension at PFP droplet-tissue interface	0.0035	N/m
ρ_2	density of PFP liquid	1590	kg/m ³
ρ_3	density of tissue	1044	kg/m ³
c_3	sound speed in the tissue	1540	m/s
κ_2	thermal conductivity of PFP liquid	0.032	W/(m K)
η_2	viscosity of PFP liquid	0.000652	Pa s
η_3	viscosity of tissue	0.005	Pa s
ε_2	specific heat of PFP liquid	1089.4	J/(kg K)
λ_3	relaxation time of tissue	3×10^{-9}	s
G	elastic modulus of tissue	2×10^5	Pa
R_{20}	initial radius of PFP droplet	1×10^{-7}	m

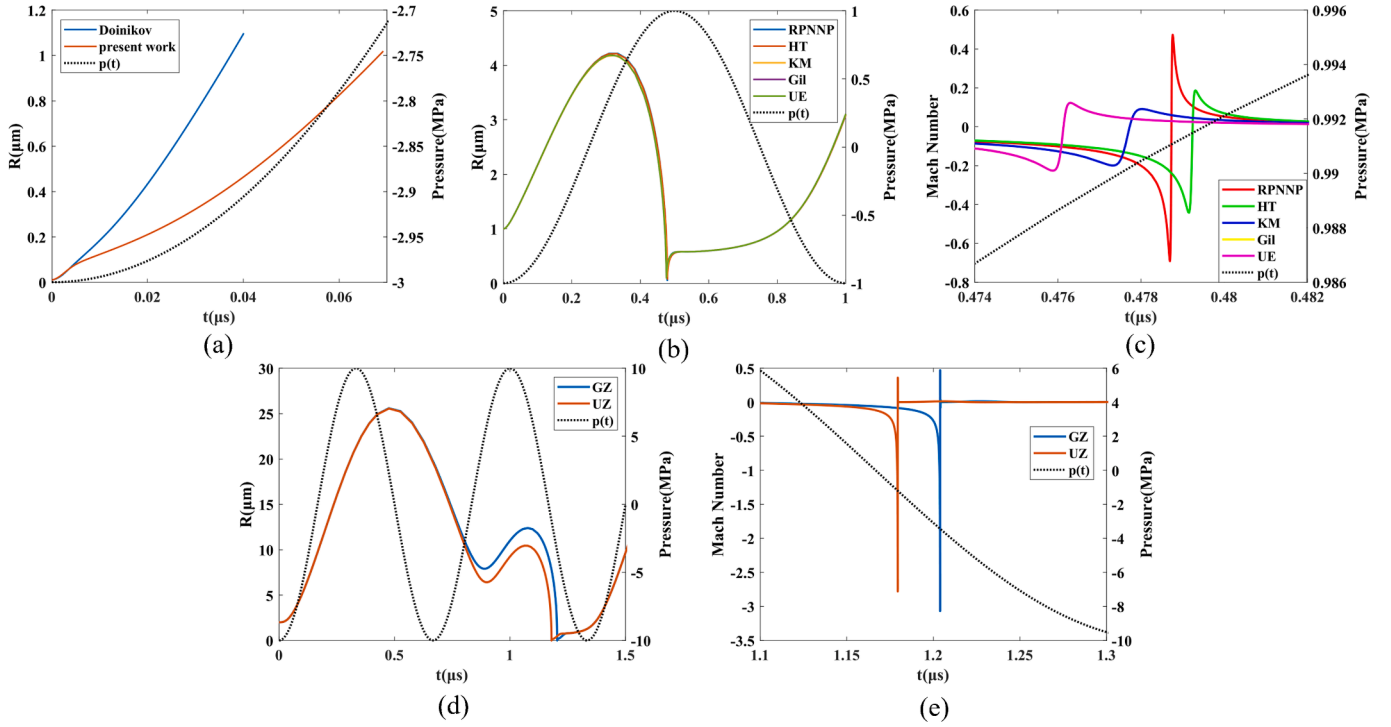


Fig. 2. (a) Comparison of the bubble vaporization model introduced by Doinikov et al. [33] and that raised in the present work. $A = 3$ MPa, $f = 1$ MHz, $R_{10} = 0.01$ μm and $R_{20} = 0.1$ μm . Comparison of the (b) bubble dynamics results and (c) Mach Number of Rayleigh-Plesset's (RPNP), Herring-Trilling's (HT), Keller-Miskis's (KM), Gilmore's (Gil) and the bubble unified equation (UE) models. $A = 1$ MPa, $f = 1$ MHz and $R_0 = 1$ μm . Comparison of the (d) bubble dynamics results and (e) Mach Number of Gilmore-Zener's (GZ) and the bubble-unified-equation-Zener (UZ) models. $A = 10$ MPa, $f = 1.5$ MHz and $R_0 = 2$ μm . Driving pressure $p(t) = -A \cos(2\pi ft)$ is applied.

excitation source. Notably, for simplification, a linear single-frequency sinusoidal pressure is used to induce ADV in this section. Apart from this, nonlinear dual-frequency waveforms, obtained from the time-domain simulation of acoustic wave fields using k-Wave, are applied to induce ADV in other sections.

In this work, we present an integrated model for bubble nucleation, growth and recondensation and oscillation in ADV. The nucleation model has been introduced and discussed in detail in our previous work [43,45]. The nucleation threshold calculated in this study is 11.2 MPa, which aligns with the value reported by Miles et al. (9.33 MPa) [41]. The discrepancy arises because we focus on different nucleation regions and timeframes when calculating the number of critical nuclei N_c . It may be higher than the experimental values reported (1.6 MPa in Ref [42]). It arises from differences in the standards used to define the threshold. In experiments, we do not specifically concern with which phase-shift droplet is triggered while the key question in this work is to determine the level of pressure required to initiate the phase transition of a phase-shift droplet at the focal point during the first pulse.

In Fig. 2(a), the comparison of the bubble growth model introduced by Doinikov et al. [33] and that raised in the present work has been shown. In this case, the radii of the initial vapor bubble and phase-shift droplet are 0.01 μm and 0.1 μm respectively. The driving pressure parameters used are $A = 3$ MPa and $f = 1$ MHz. The main difference between the two models is that the environment around the droplet is different. In the Doinikov et al.'s model, the host liquid is assumed as Newtonian fluid and in our present model, the phase-shift droplet are distributed in biological tissue which is modelled as Zener viscoelastic model. Considering the viscoelasticity of the tissue, the growth process of vapor bubble is constrained with lower speed and similar but smaller size after vaporization (1.02 μm in our model and 1.10 μm in the Doinikov et al.'s model).

In Fig. 2(b) and Fig. 2(c), the viscoelastic tissue model was ignored and Mach Numbers remained low due to the low excitation condition (A

$= 1$ MPa, $f = 1$ MHz and $R_0 = 1$ μm). In this case, the performances of different models were similar and significant differences are only observable during bubble collapse. Gilmore model is the most widely used model for bubble dynamics in recent years for its accuracy and wide application range. As shown in Fig. 2(b) and Fig. 2(c), there was minimal difference between the Unified Equation (UE) for bubble dynamics and Gilmore (Gil) models in describing bubble oscillation under low intensity excitation source. The comparison of the results of UE and Gil models considering the viscoelastic Zener model under high excitation condition ($A = 10$ MPa, $f = 1.5$ MHz and $R_0 = 2$ μm) was demonstrated in Fig. 2(d) and Fig. 2(e). Under such a condition, the Mach Number exceeds 2.2 which surpasses the applicability of Gilmore model while the unified theory for bubble dynamics is not subject to this limitation [30]. The model employed in this paper proves to be more universally valid and effective, particularly under high acoustic excitation conditions considering the viscoelasticity of the tissue.

3.2. Nucleation onset time and nucleation size

The time required to generate a critical nucleus at LF = 0.5 MHz and HF = 3 MHz is illustrated in Fig. 3(a). The regions without data indicate no nucleation occurred within 3 s of irradiation and a zero value signifies that nucleation occurred immediately after irradiation. As the acoustic power increases, nucleation onset time decreases, indicating that a lower temperature is required at higher acoustic pressures as temperature rises with irradiation time. The time-dependent temperature variation is detailed in our previous work [43,45].

Fig. 3(b) illustrates the radius of the critical nucleus generated under the conditions described in Fig. 3(a). As shown in Eq. (1), it is expected to see a negative correlation between the size of the critical nucleus and acoustic power as higher acoustic power results in higher negative pressure (It should be pointed out that the negative pressure and negative stress mentioned in this article refer to their absolute value.).

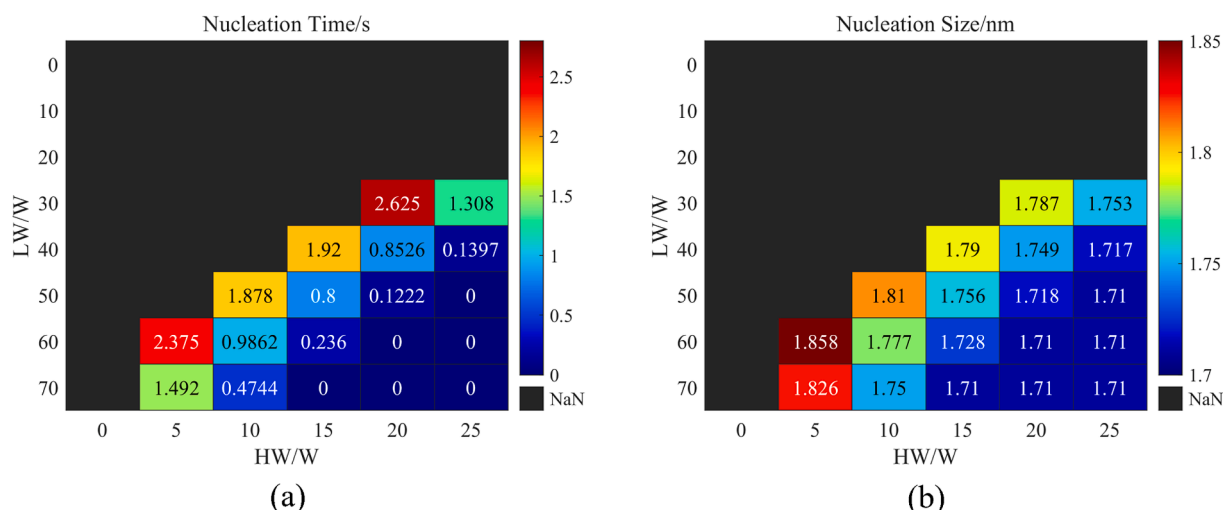


Fig. 3. (a) Minimal irradiation time needed to generate nucleation. (b) The corresponding radius of the critical nucleus. LF = 0.5 MHz and HF = 3 MHz.

As shown in Fig. 3(b), the initial radii of the critical nuclei are closely clustered around 1.7 nm. The reason may be that the number of critical nuclei is the same in all cases (i.e., equal to 1). In particular, when the time required in Fig. 3(a) is zero, the external conditions for forming a single critical nucleus are nearly identical.

3.3. Vaporization dynamics of PFP nanodroplet

The whole-process vaporization dynamics of the PFP nanodroplet is shown in Fig. 4, with time starting after nucleation is complete. In Fig. 4(a), the critical nucleus failed to grow into a vapor bubble and instead condensed back into nanodroplet. The dual-frequency ultrasound parameter for this case was LF = 0.5 MHz, LW = 60 W, HF = 3 MHz and HW = 5 W. It demonstrates that the critical nucleus is unstable and naturally tends to condense [44]. By increasing the excitation frequency and acoustic power, the resulting high negative pressure can counter-balance the natural condensation, facilitating the growth of vapor bubble. Therefore, in the cases analyzed in this study, the majority of phase-shift droplets did not undergo the condensation process. For the sake of clarity, we will refer to the second process of ADV as the growth process instead of growth and recondensation process in the subsequent text.

Fig. 4(b) and Fig. 4(c) present the whole-process vaporization dynamics for the case of LF = 1.5 MHz, LW = 40 W, HF = 3 MHz and HW = 15 W. As shown in Fig. 4(b) and Fig. 4(c), with the enhancement of excitation energy, the vaporization process is completed and the

moment when nanodroplet completely turns into vapor is marked as a cross. Both the growth and oscillation process are strongly influenced with the excitation acoustic pressure so the dual-frequency ultrasound waveform is also displayed as dashed lines in the figures. Fig. 4(b) shows the simulated results of vapor radius within one ultrasonic cycle and Fig. 4(c) presents the corresponding Mach Number results. The initial radius of the critical nuclei is 1.71 nm and by the end of vaporization, the radius of vapor bubble grows to 4.96 μm . The droplet undergoes significant expansion in the growth stage, and the validity of this value will be discussed in the following section. The maximum radius during oscillation reaches 21.73 μm , with an expansion ratio of 4.38. Many related studies have reported comparable results regarding the expansion ratio [12,19,33,60]. The maximum Mach Number is up to 2.56, indicating a significant change during bubble collapse. It also confirms that under such conditions, the commonly used Gilmore model is inapplicable, highlighting the importance of employing the unified bubble dynamics equation along with the viscoelastic Zener model.

3.4. Effects of dual-frequency ultrasound parameters on the bubble radius during the whole process

The bubble radius at the completion of the growth process serves as the initial radius for the oscillation process. The variation of the post-growth vapor radius with frequency and acoustic power is presented in Fig. 5. As can be seen in Fig. 5 that the bubble radius after growth is relatively large compared to the initial nanodroplet radius. The main

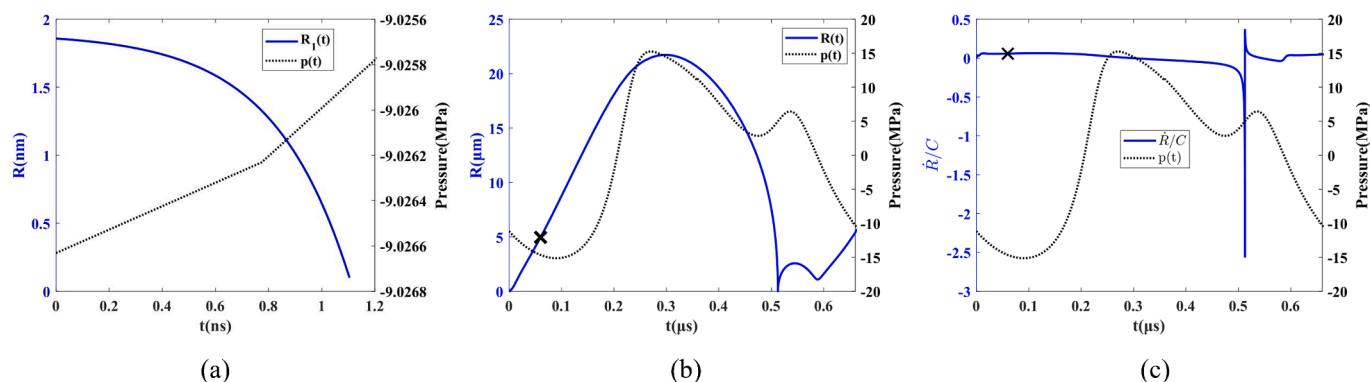


Fig. 4. (a) The time-dependent changes in the vapor bubble radius for a case of condensation. LF = 0.5 MHz, LW = 60 W, HF = 3 M and HW = 5 W. (b) The time-dependent changes in the vapor bubble radius for a case of growth and (c) the corresponding results of Mach Number. LF = 1.5 MHz, LW = 40 W, HF = 3 MHz and HW = 15 W.

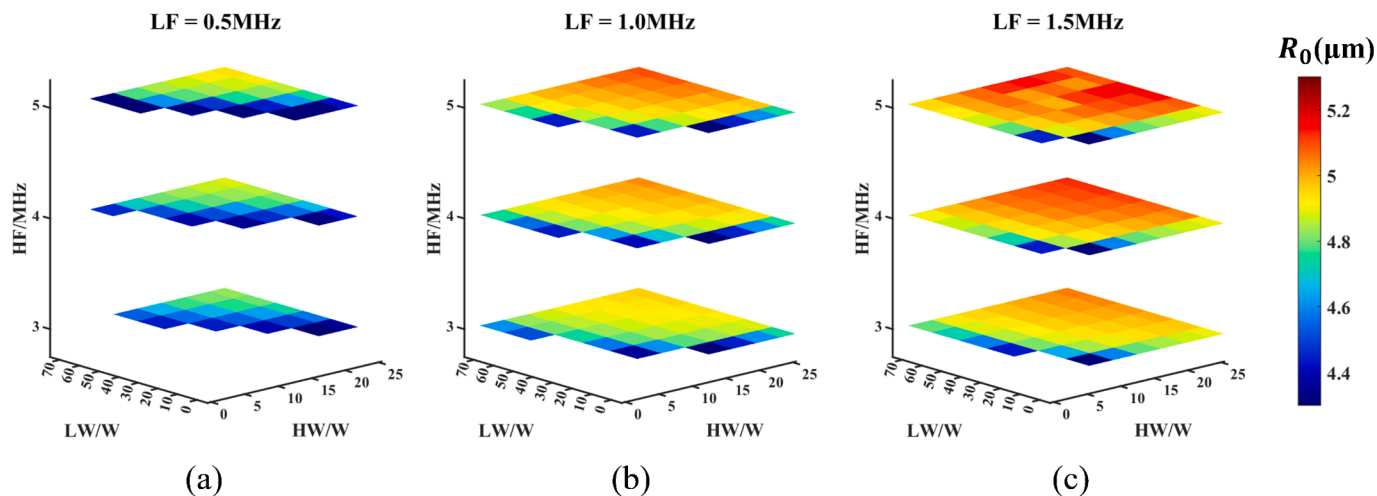


Fig. 5. The post-growth vapor radius under different dual-frequency focused ultrasound conditions.

reason is that the density difference between the liquid and gaseous phases is substantial. In addition, due to the dual-frequency ultrasound is introduced, the driving pressure applied in our study (approximately 10 MPa) is significantly higher than that in related work [35]. If we decrease the pressure amplitude like what we did in Fig. 2(a), the result will be consistent with the previous work. Consequently, such an expansion of post-vaporization radius does not contravene the law of conservation of mass. In most cases, the post-growth vapor radius increases with increasing ultrasound frequency and acoustic power. As shown in Fig. 4(b), the growth process is very rapid, completing within 0.1 μ s. During this period, the waveform of $p_{ac}(t)$ follows a simple pattern without any significant incremental or subtractive changes. In addition, the radii of the critical nuclei are similar, they determine that the values of the post-growth vapor radius are concentrated around 5 μ m. Some exceptions could be observed in Fig. 5(c) when LF = 1.5 MHz and HF = 5 MHz. The reason might be that with the increase of frequency, the waveform of $p_{ac}(t)$ reaches an extreme point, altering its monotonicity [61]. However, these exceptions do not affect the overall trend.

Fig. 6 illustrates the maximum radius during the whole-process vaporization dynamics within one ultrasonic cycle under different dual-frequency focused ultrasound conditions. Similarly, the maximum radius generally exhibits a positive correlation with acoustic power in most cases. However, the effect of frequency on the maximum radius is more complex. In general, the maximum radius increases as the LF

increases, while the relationship with HF remains unclear. Importantly, HF and LF should not be considered independently, as their combination substantially influences the maximum radius due to variations in waveform and phase [62]. The crests and troughs of the waveforms correspond to alternating expansion and contraction of the vapor bubble. In addition, dual-frequency ultrasound parameters also indirectly influence the maximum radius through the initial radius (i.e., the post-growth vapor radius, as shown in Fig. 5). Previous research has demonstrated that the maximum radius initially increases and then decreases with an increase in the initial radius [63]. Thus, two distinct nonlinear influence mechanisms are suggested, making it challenging to establish a straightforward relationship between the maximum radius and dual-frequency ultrasound parameters.

3.5. Effects of dual-frequency ultrasound parameters on the mechanical stress during the whole process

The maximum absolute values of Mach Number during one ultrasound cycle under different dual-frequency focused ultrasound conditions are shown in Fig. 7. Firstly, it can be observed that in most cases, the Mach Number exceeds 2.2, which goes beyond the valid application range of the commonly used Gilmore model [27]. It suggests that the inclusion of high-frequency ultrasound could enhance the intensity of the bubble collapse. It should be pointed out that one may read the results from related work that the expansion of ADV bubbles are

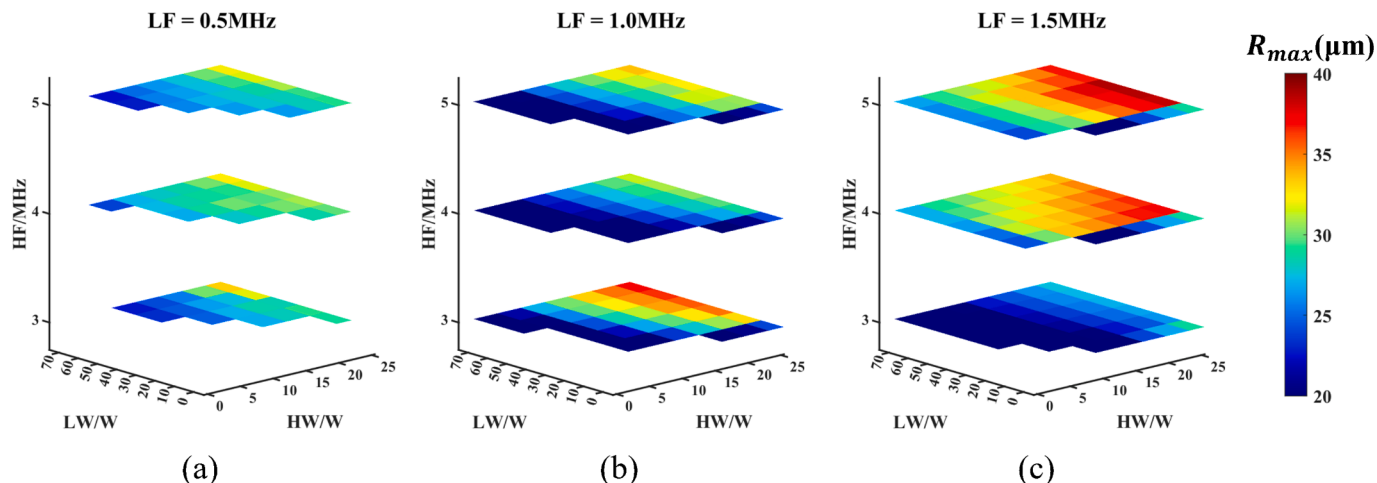


Fig. 6. Maximum radius within one ultrasound cycle under different dual-frequency focused ultrasound conditions.

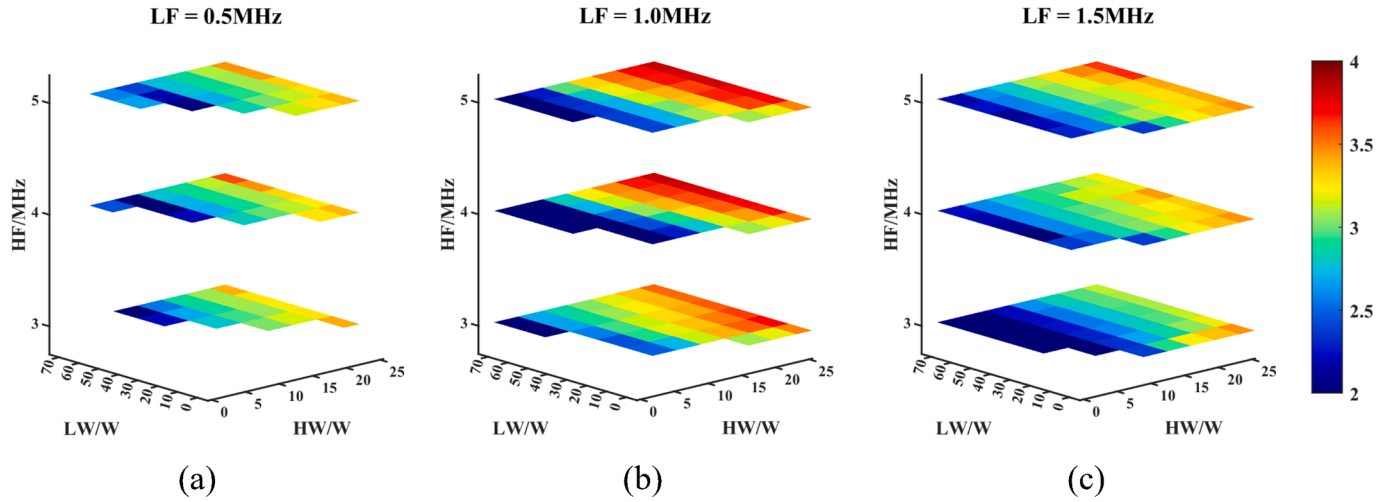


Fig. 7. Maximum absolute values of Mach Number within one ultrasound cycle under different dual-frequency focused ultrasound conditions.

significantly lower at higher frequencies [60]. The perceived contradiction arises from the differing variables held constant when examining the effects of driving frequency. In Ref [60], focal pressure on ADV bubbles was maintained consistently. In contrast, our study emphasizes the ultrasonic driving system, including the waveform generator and radiofrequency amplifier, by maintaining consistent acoustic power. This approach is more practical for real-world applications, particularly under dual-frequency ultrasound conditions and in the complex propagation media of ultrasound waves. With consistent acoustic power, the pressure increases with driving frequency, as demonstrated in Fig. S1 and Fig. S2 in the Supplementary Material.

Additionally, as expected, higher acoustic power generally results in a higher Mach Number. However, the effects of low-frequency and high-frequency ultrasound on the Mach Number is different. Generally speaking, the Mach Number increases with higher HF while it initially increases and then decreases with higher LF. As LF increases, the nonlinear distortion of ultrasonic waveform becomes severer, causing the bubble to collapse multiple times within a single ultrasound cycle (as also illustrated in Fig. 9(b) below). With an increased number of collapses, the intensity of each collapse weakens.

Fig. 8 demonstrates the spatiotemporal distribution of the stress τ_3 (Fig. 8(a)) and strain γ_3 (Fig. 8(b)) applied to the tissue under the condition of LF = 1 MHz, LW = 40 W, HF = 4 MHz and HW = 15 W. As shown in Fig. 8(a), during the expansion phase, the bubble (or the

droplet before the growth stage is complete) compresses the surrounding tissue, resulting in compressive (negative) stress. During the contraction phase, as the deformed tissue returns to its initial state, significant tensile (positive) stress is generated. It is important to note that since the viscoelastic tissue is modeled using the Zener model, an ordinary differential equation over time, the negative and positive stress do not correspond to the expansion or contraction of the bubbles in real-time [51].

However, according to Eq. (20), it can be seen from Fig. 8(b) that the magnitude and sign (positive or negative) of the strain depend directly on the size of the bubble. Considering the viscoelasticity of tissue, it possesses memory and recovery characteristics, which stress can capture more effectively than strain. In addition, while the strain generated in water and tissue is similar [64], tissue's higher viscosity and elasticity mean that the same strain generates significantly larger stresses in tissue. And it is stress, rather than strain, that directly causes damage to cells. Therefore, the stress could better reflect the mechanical effects during the whole ADV process, and in the following sections, we will mainly discuss stress rather than strain.

The absolute value of positive stress is an order of magnitude larger than that of negative stress. It could be deduced through straightforward calculations by Eqs. (19–21). On the right-hand side of Eq. (19), the order of magnitude of $2G\gamma_{3,r}$ is approximately 10^5 , whereas that of $2\eta_{3,r}\dot{\gamma}_{3,r}$ is approximately $10^6 \sim 10^7$. Thus, the strain rate has a greater

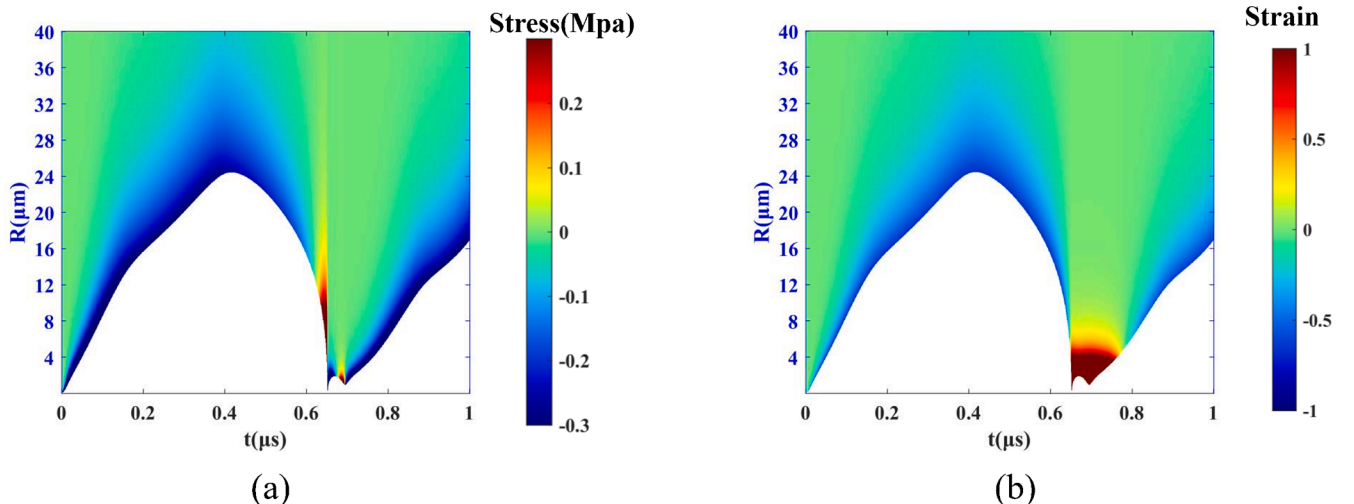


Fig. 8. The spatiotemporal distribution of the (a) stress and (b) strain applied to the tissue. LF = 1 MHz, LW = 40 W, HF = 4 MHz and HW = 15 W.

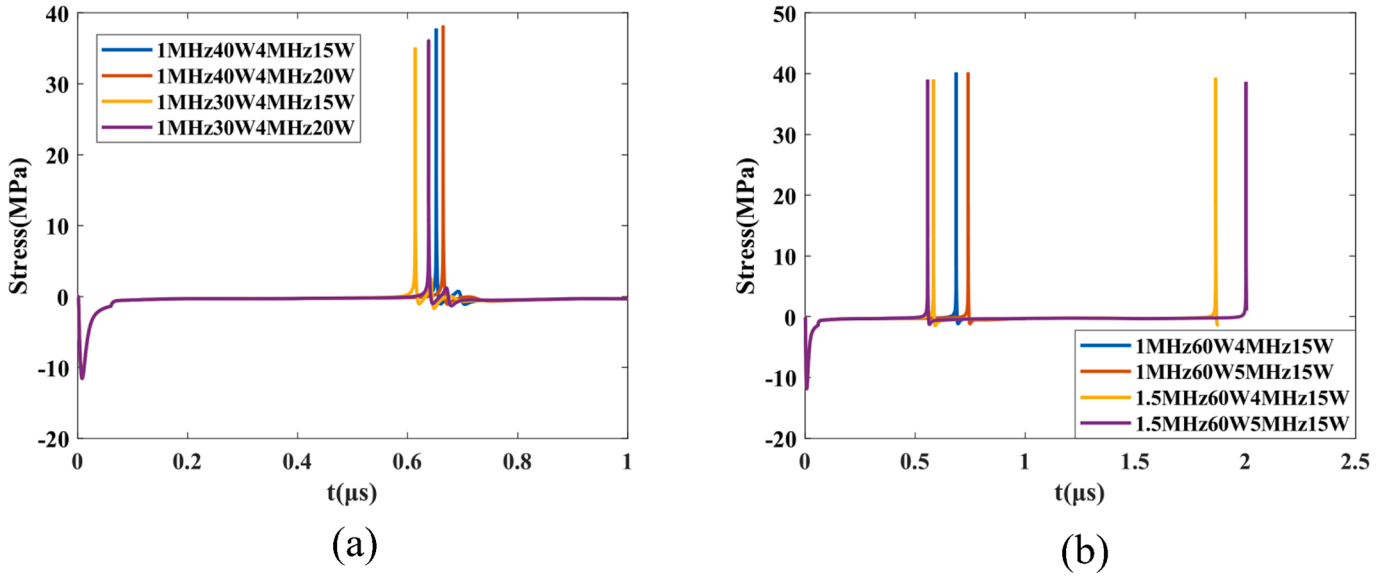


Fig. 9. The effects of (a) acoustic powers and (b) frequency on the values of stress at the interface of tissue and bubble (or droplet).

impact on stress than strain. When bubbles shrinkage and collapse, the strain rate of the tissue is greater than when they expand. And the maximum strain rate occurs during bubble collapse; consequently, the maximum stress also occurs during bubble collapse, where it manifests as tensile stress. This explains why the tensile stress is significantly higher than the compressive stress.

To effectively demonstrate the spatial distribution of τ_3 , appropriate color bar range was selected in Fig. 8(a). The stress values at the tissue-bubble (or droplet) interface during one ultrasound cycle are presented in Fig. 9(a) and Fig. 9(b). Fig. 9(a) presents the effects of acoustic power, while Fig. 9(b) illustrates the effects of frequency. Increasing acoustic power leads to higher positive stress, whereas changes in frequency can alter the oscillation mode due to the nonlinear distortion of the ultrasonic waveform. The peak stress coincides with the moment of maximum Mach Number, i.e., when the bubble collapses. Therefore, the Mach Number can also indicate the mechanical effects during ADV.

Fig. 10 depict the maximum values of stress during one ultrasound cycle under different dual-frequency focused ultrasound conditions. In this study, the maximum stress across all cases is approximately 45 MPa. This is consistent with previous related research [64] and falls within a reasonable stress range. The variation trends of the values with dual-frequency ultrasound parameters are similar to that shown in Fig. 7.

As presented in Fig. 8 and Fig. 9, the maximum value of stress occurs during bubble collapse, which coincides with the moment when the Mach Number reaches its peak. To avoid redundancy, it will not be further discussed here. However, it is important to note that the effect of HF on stress is less pronounced compared to its effect on the Mach Number.

During the initial growth stage, the droplet expands rapidly, generating significant negative stress on the surrounding tissue [33]. However, this occurrence is brief and limited in duration. As a result, it is not clearly visible in Fig. 8(a), but it can be easily observed in Fig. 9.

3.6. Effects of dual-frequency ultrasound parameters on the heat deposition during the whole process

Three types of heat deposition during one cycle under the condition of LF = 1 MHz, LW = 40 W, HF = 4 MHz, HW = 15 W and $r' = 0.1$ mm are shown in Fig. 11. It can be observed that Q_{ac} fluctuates with ultrasound wave, whereas both Q_{vis} and Q_{rad} exhibit a peak only during bubble collapse. Additionally, the magnitudes of the three heat deposition types vary significantly, with Q_{vis} being the biggest and Q_{ac} the smallest among them. Both Q_{ac} and Q_{rad} are related to the attenuation term while Q_{vis} is associated with the viscosity term [57]. The results

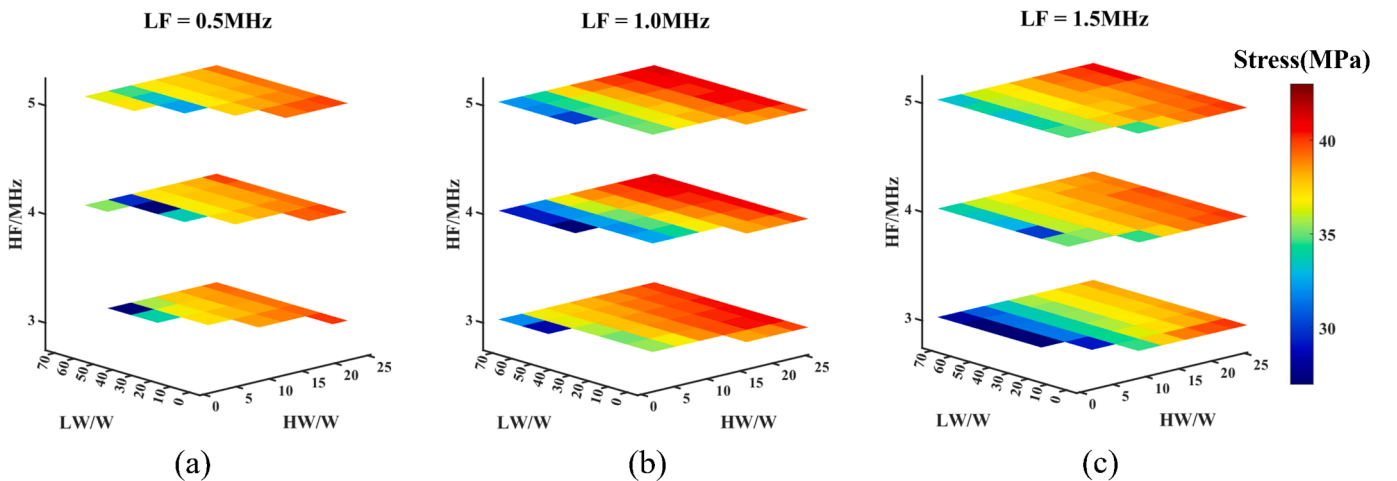


Fig. 10. Maximum values of stress within one ultrasound cycle under different dual-frequency focused ultrasound conditions.

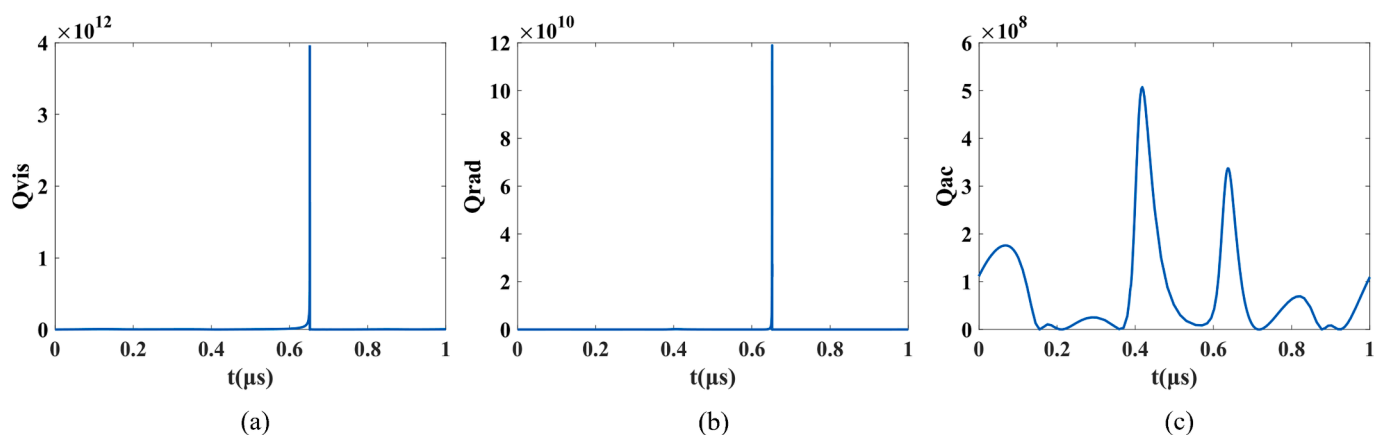


Fig. 11. Three types of heat deposition during one ultrasound cycle.

indicate that the viscosity term contributes far more to the heat deposition during the bubble oscillation particularly at bubble collapse.

The total heat deposition can be calculated by $Q = Q_{ac} + Q_{rad} + Q_{vis}$. Fig. 12 illustrates the trends of maximum values of Q over one cycle as they vary with dual-frequency ultrasound acoustic powers and frequencies. As previously discussed, Q is primarily dominated by Q_{vis} which peaks during bubble collapse. As expected, the effects of dual-frequency ultrasound parameters on the maximal Q are similar to that on maximal stress and maximal Mach Number, as all these peaks occur at bubble collapse. Detail descriptions of the effects could be found in the preceding sections.

4. Discussion

This work aimed to investigate the effects of dual-frequency ultrasound parameters on the whole-process vaporization dynamics of a nanodroplet inside a viscoelastic medium. A numerical model combining fluid dynamics and thermodynamics methods was developed to simulate the acoustic vaporization of a PFP nanodroplet. The diameter of critical nucleus within the nanodroplet was determined using modified classical nucleation theory based on the temperature and pressure simulations of nonlinear propagations of dual-frequency ultrasound. After vaporization, the oscillation of the vapor bubble was described using the unified theory for bubble dynamics. The Zener viscoelastic model was employed to simulate the characteristics of soft tissue.

The entire vaporization process, including nucleation, growth and

oscillation stages, is initiated by a continuous waveform, making the process continuous and successive. When the negative pressure increases to make the pressure inside the nanodroplet p_l exceeds the nucleation threshold, a critical nucleus will be formed [65]. The growth stage begins immediately thereafter, with the waveform continuing from the point where nucleation is completed. Since p_l can trigger nucleation, it can also counterbalance the natural condensation of the vapor bubble, aiding its growth in most cases. The occurrence of condensation is rare and one example of which is shown in Fig. 4(a). Similarly, high peak driving acoustic pressure also results in direct growth of vapor bubble without any rebound, as observed in other studies [15,35]. As shown in Fig. 7, the Mach numbers at bubble collapse under dual-frequency ultrasound excitations are significantly higher, resulting from increased acoustic pressure compared to traditional single-frequency ultrasound. This study has confirmed the theory's consistency with existing models under low-amplitude excitation and demonstrated its robustness under high-amplitude conditions. Our work aims to expand the scope of biomedical research on bubble collapse phenomena, particularly in scenarios involving high Mach Numbers.

Previous researchers have traditionally focused on the aforementioned stages in isolation to examine the effect of acoustic amplitude. In contrast, our simulation integrates all stages to provide a more comprehensive and scientifically accurate model of the entire ADV process. Our whole-process model emphasizes the continuity throughout the entire process. After nucleation, the critical nucleus size becomes the initial radius within the nanodroplet during vapor growth. Meanwhile, the size of the post-growth vapor defines the initial radius

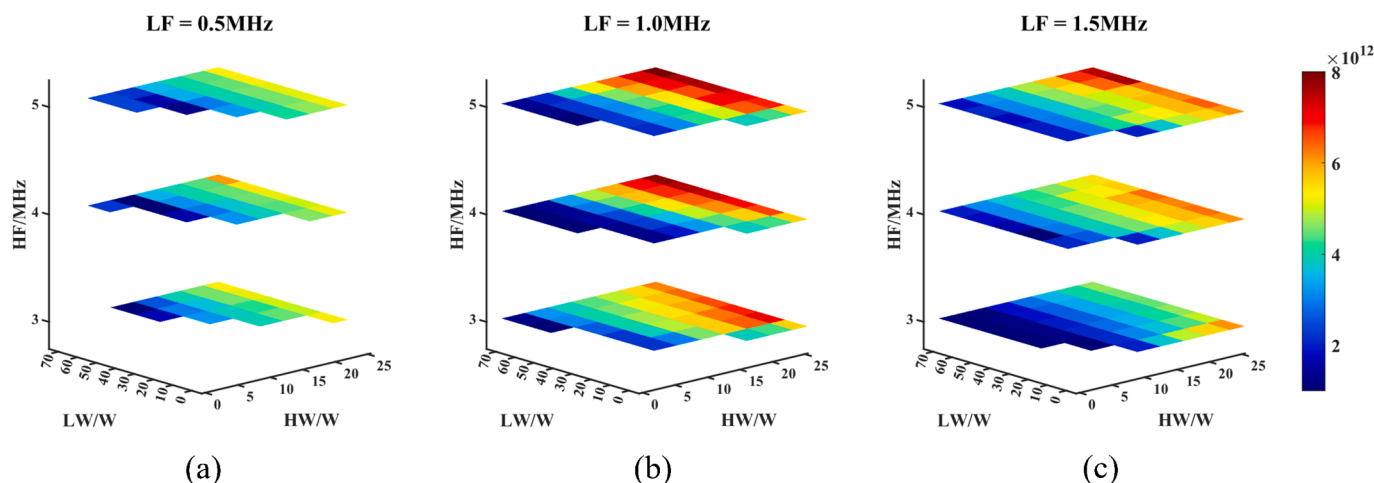


Fig. 12. Maximum values of heat deposition within one ultrasound cycle under different dual-frequency focused ultrasound conditions.

for bubble dynamics. In contrast, in previous staged models, the initial radius is often assumed to be constant and independent of ultrasound excitation conditions. Additionally, the entire process in our model is stimulated by a continuous ultrasound signal, and the phase of the ultrasound wave plays a critical role. In previous staged models, the initial phase is often set at the moment when negative pressure is at its maximum, to ensure that the phase transition process is as efficient as possible. In contrast, our model initiates when nucleation is complete. When the onset time of nucleation, as shown in Fig. 3(a), is non-zero, the waveform starts at the peak negative pressure, similar to that of staged models. In other cases, the process begins at when p_- is 11.2 MPa, the pressure required to form a critical nucleus at room temperature. The variation in the initial phase significantly affects subsequent bubble dynamics, which may also explain the complex pattern of R_{max} under dual-frequency ultrasound parameters, as shown in Fig. 6.

In previous studies, some researchers have suggested that larger bubbles cause more severe damage during acoustic cavitation and there is a proportional relationship between them [19,66,67]. However, it cannot be deduced from our results. Our simulations indicate that the cavitation-associated damage, including stress and heat, is proportional to the maximum Mach Number during the bubble collapse. Previous researchers may have reached their conclusions by neglecting the nonlinear propagation of dual-frequency ultrasound. Nonlinearity can be ignored only when the acoustic pressure is relatively low, under which conditions bubble oscillations are less intense, and waveform distortion has minimal impact [36]. In such cases, the mode of bubble dynamics is simple and one may observe a positive correlation between high Mach numbers and large bubble radii [67], which may have misled earlier studies. Fundamentally, the cavitation-associated damage is linked to the rapid collapse of the bubble which produces intense mechanical and thermal damages rather than the maximum bubble size. As shown in Fig. 8, Fig. 9 and Fig. 11, the maximum stress and heat deposition occur at the moment of collapse instead of the moment when the bubble reaches its maximum size.

In this work, nano-sized phase-shift droplets are applied which are capable of passing through the blood vessel wall to reach the tissue and phase-shift droplets are distributed within a tissue phantom simulated with a viscoelastic Zener model. Compared to microdroplets which are traditionally used for embolotherapy [68], there is a significant difference in the calculated ADV-induced stress caused to the surrounding medium. Tissue is more viscous and elastic than blood, therefore, in tissue, the same strains generate larger stresses. Such an intense stress could potentially be applied in bubble-enhanced histotripsy and lithotripsy.

Although both mechanical and thermal damages to ambient tissue are directly related to bubble collapse, there is a notable difference in how dual-frequency ultrasound parameters affect each. As shown in Fig. 9 and Fig. 10, an increase in HW significantly amplifies the maximum stress. In contrast, Fig. 12 demonstrates that adjusting the frequency provides a more effective way to regulate heat deposition. This difference may be attributed to the fact that, unlike stress, heat deposition is also influenced by the absorption properties of the surrounding tissue, which are frequency-dependent according to power-law acoustic absorption. In practical applications, mechanical damage can be intensified by increasing HW while keeping thermal damage relatively constant. This approach could be valuable in guiding the selection of ultrasound parameters for procedures such as thrombolysis or lithotripsy, where the mechanical effects of ultrasound are leveraged while minimizing thermal harm to surrounding blood vessels and other tissues.

An interesting phenomenon observed in the simulation results is that the maximum values for bubble radius, Mach number, stress, and heat deposition under the dual-frequency combination of 1.5 MHz + 3 MHz are abnormally smaller compared to the general trend with dual-frequency ultrasound parameters. This may be attributed to the nonlinearity of dual-frequency ultrasound. As shown in our previous

paper [45], the combination of fundamental and second harmonic like 1.5 MHz + 3 MHz exhibits unique characteristics. These include a relatively lower rp (the ratio of the values of peak negative pressure of nonlinear model to that of linear model) and a higher DoN (degree of nonlinearity, $\text{DoN} = A_{f=LF+HF}$). A lower rp indicates a reduced peak negative pressure and a higher DoN suggests fewer higher harmonics component since more energy is concentrated at the sum frequency. Both factors contribute to the abnormally smaller simulation results because the process of bubble dynamics is relatively mild under lower pressure and frequency conditions. This can prolong the lifespan of bubbles [69], potentially enhancing the efficacy of ultrasound treatment. This might explain why such dual-frequency combinations are increasingly used in recent studies.

It is notable that the ADV nucleation, simulated by CNT, strongly depends on macroscopic parameters such as temperature and pressure. However, it is reported that other mechanisms, such as droplet deformation or hydrodynamic cavitation [1,70,71], can also influence ADV nucleation and are not accounted for in CNT. This may result in discrepancies between the predictions made by CNT and those obtained from experiments. It should also be pointed out that we did not account for the effects of adjacent droplets or bubbles on the vaporization dynamics. Previous studies suggest that adjacent droplets or bubbles may suppress the vaporization process [35,72]. This effect should be considered in our future work.

5. Conclusions

In this work, we developed a comprehensive model for whole-process vaporization dynamics which unified nucleation, growth, and oscillation stages and associated bioeffects of nonlinear dual-frequency excitation in a viscoelastic medium. Parametric analyses were conducted to investigate the influence of dual-frequency ultrasound frequencies and acoustic powers on the bubble radius, Mach Number, stress and heat deposition. Our simulations demonstrated that the model accurately captured the whole process of ADV, with Mach Number exceeding 2.2 due to the nonlinear effects of dual-frequency ultrasound. Moreover, the results indicated that rapid bubble collapse, rather than maximum bubble size, was directly linked to cavitation-induced mechanical and thermal damage. We also found that mechanical and thermal effects could be independently regulated to some extent, given their differing sensitivities to frequency and acoustic power. Notably, certain dual-frequency combinations, such as 1.5 MHz + 3 MHz (fundamental and second harmonic), deviated from general trends, likely due to a higher DoN and lower rp , which results in reduced bubble radius. This phenomenon may extend bubble lifespan, potentially enhancing the efficacy of ultrasound treatments. In conclusion, our model offers a more comprehensive and scientifically grounded representation of whole-process ADV which could be applied in high intense ultrasound conditions. This work advances our understanding of the complex damage mechanisms in nanodroplet-enhanced ultrasound therapy, contributing to its practical clinical application.

CRedit authorship contribution statement

Yubo Zhao: Writing – original draft, Software, Resources, Methodology, Conceptualization. **Yi Feng:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Liang Wu:** Writing – review & editing, Validation, Supervision, Software, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ultsonch.2025.107234>.

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